

Midterm Lab Quiz 2: Advanced String Manipulation and Hashing

1. The program should accept four distinct string parameters Barangay, City, Province and Country
2. String Extraction: Extract the first two characters of the Barangay and City and the last two characters of the province and Country.
3. Generate a Keccak-256 hash of the combined result using both `abi.encode` and `abi.encodePacked` methods.
4. Generated outputs:
 - Display the individually extracted characters.
 - Display the newly generated concatenated string.
 - Ensure the Full Name of the user is included as part of the contract's generated output or state metadata.

Code:

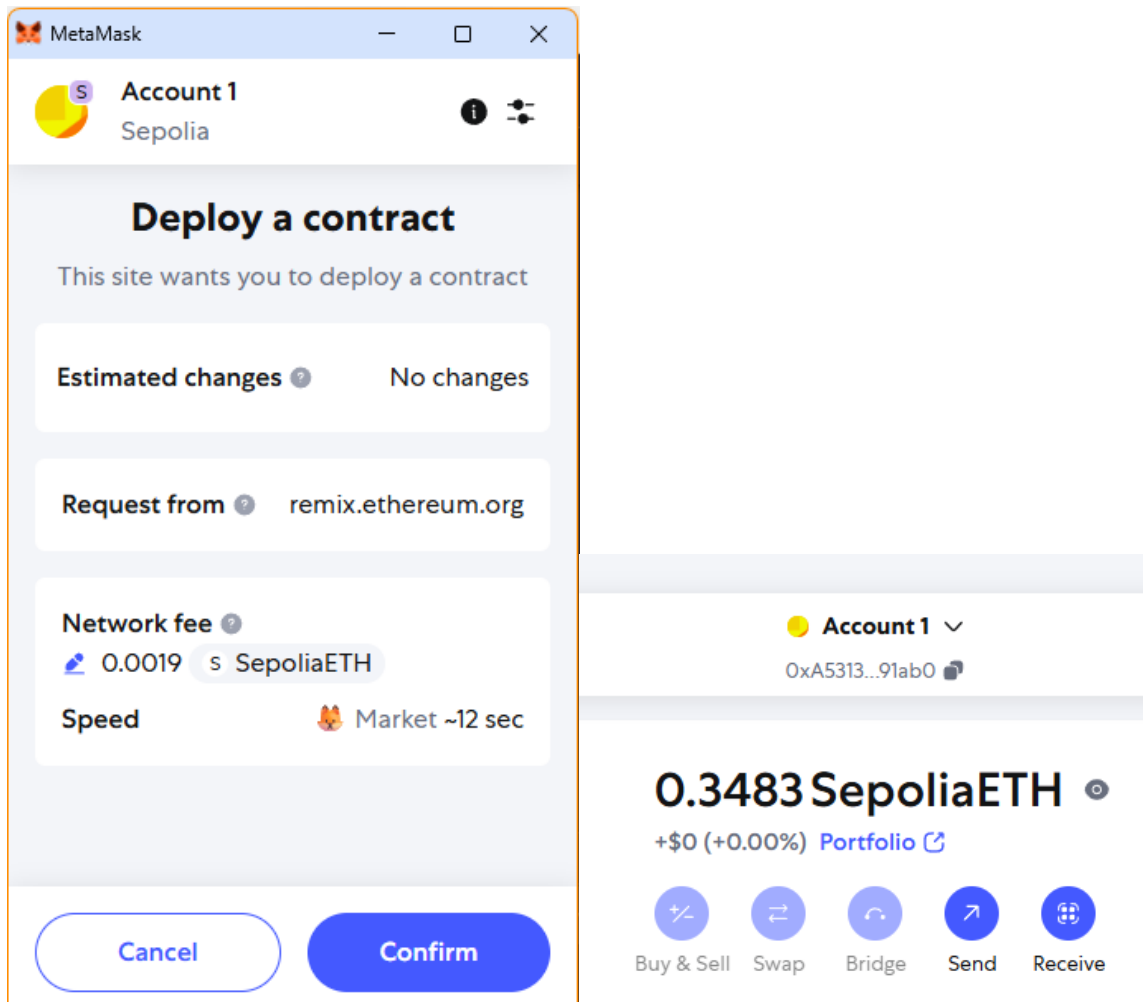
```

1 // Opiniano, Ezkylle
2 // NW - 301
3
4 // SPDX-License-Identifier: MIT
5 pragma solidity ^0.8.20;
6
7 contract SimpleStringHash {
8
9     // User metadata
10    string public fullName;
11
12    // Outputs
13    string public extracted;
14    bytes32 public hashEncode;
15    bytes32 public hashPacked;
16
17    constructor(string memory _fullName) {  infinite gas 587600 gas
18        fullName = _fullName;
19    }
20
21    function process(  infinite gas
22        string memory barangay,
23        string memory city,
24        string memory province,
25        string memory country
26    ) public {
27        // Convert to bytes for indexing
28        bytes memory b1 = bytes(barangay);
29        bytes memory b2 = bytes(city);
30        bytes memory b3 = bytes(province);
31        bytes memory b4 = bytes(country);
32
33        // Extract required characters
34        string memory result = string(
35            abi.encodePacked(
36                b1[0], b1[1], // first 2 of Barangay
37                b2[0], b2[1], // first 2 of City
38                b3[b3.length - 2], b3[b3.length - 1], // last 2 of Province
39                b4[b4.length - 2], b4[b4.length - 1] // last 2 of Country
40            )
41        );
42
43        extracted = result;
44
45        // Hashing
46        hashEncode = keccak256(
47            abi.encode(result)
48        );
49
50        hashPacked = keccak256(
51            abi.encodePacked(result)
52        );
53    }
54 }
55

```

Screenshots:

Before Deploying:



After Deploying:

▼ SIMPLESTRINGHASH AT 0X8E5

Balance: 0 ETH

PROCESS

barangay: Telebastagan

city: SanFernando

province: Pampanga

country: Philippines

Calldata

Parameters

transact

extracted

0: string: TeSagaes

fullName

0: string:

hashEncode

0: bytes32: 0x35cbc07f7189f946719fc6b0a62ccb3cde7290519b86ccda
fb2068b264eeddc7

hashPacked

0: bytes32: 0x278082b943a9737eb0f7d69aef862cd6ea6aa51be55a404
5401259cab9828f85

Account 1 ▼

OxA5313...91ab0

0.3463 SepoliaETH

+\$0 (+0.00%) Portfolio

Buy & Sell

Swap

Bridge

Send

Receive